

Data Centers Industry Pack

The Data Centers Industry Pack is a collection of resources that provide pre-built, ready to use solutions for data centers. This includes faceplate & UDT pairs for an Interruptible Power Supply (UPS) and an Air Handling Unit (AHU).

This pack also includes several widgets, and a dynamic popup for displaying tag names and values.

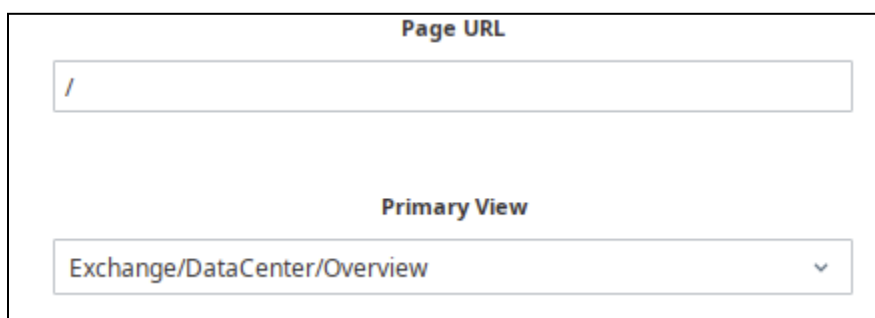
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Getting Started

Welcome to the Data Centers Industry Pack.

1. Import the **data-centers-industry-pack.zip** file.
This file can be imported through the Ignition Designer or the Gateway Config > Ignition Exchange webpage, as detailed here:
<https://docs.inductiveautomation.com/docs/8.1/platform/Gateway/ignition-exchange>
2. After importing the .zip file, review the [Tags and UDTs → Setup](#) section below for importing the UDTs and some configuration requirements for using the industry pack resources.
3. Map **Exchange/DataCenter/Overview** view to a desired URL.



The screenshot shows a configuration window with two main sections. The top section is labeled "Page URL" and contains a text input field with a forward slash "/" inside. The bottom section is labeled "Primary View" and contains a dropdown menu with "Exchange/DataCenter/Overview" selected and a downward arrow on the right.

4. (Optional) Note that the Industry Pack was designed with Ignition's dark theme in mind, but the industry pack will use your Designer's default theme. Everything will work as intended in light theme, but if you prefer to change the theme to dark mode, go to the sessions properties, and find the property named **theme**.
5. (Optional) Note that this resource includes an optional advanced **stylesheet.css** file to implement. This adds one visual adjustment for the Comtrade resource only. This is optional and can be ignored altogether if you are not using the Comtrade resource. Implementing this will:
 - Remove the area-portion of the Time Range Selector component.

If your project is already using an advanced stylesheet, you can append the code within the stylesheet.css file to the existing stylesheet. If there is no advanced stylesheet it should be enabled first.

See the Ignition docs page for more information about using/enabling the advanced stylesheet:

<https://docs.inductiveautomation.com/docs/8.1/ignition-modules/perspective/styles/style-classes#enabling-the-advanced-stylesheet>

Overview



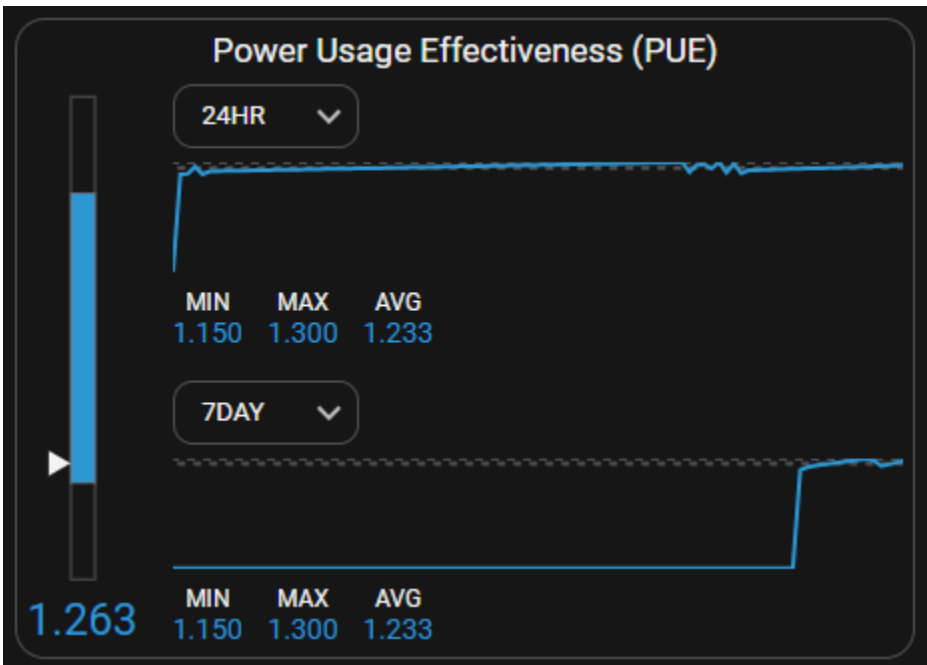
Each of the widgets and faceplates can be viewed on the Overview page.
Exchange/DataCenter/Overview

The idea here is to show everything that's included with the pack. Launching this page in a Perspective session can be used as a small demo of the pack. The widgets and popups on this view are connected to UDTs and simulated tags so the resources and all of their tag bindings work and display real-time data when it's opened. We display our widgets on the top, and buttons to open the faceplates and popups on the bottom. Each element includes a brief description here as well.

For the sparkline historical trends, it will take some time for history to be generated when installing the pack initially.

Widgets

KPI Detail Widget



This **KPI Detail Widget** widget is pointed at a single tag, which then displays the real-time value in a moving analog indicator to provide context of where the real-time value lies within a desired or expected range.

As long as the tag used with the widget is historized, sparkline trends will then be displayed showing 24 hour and 7 day history to show at a glance what's been happening recently with that particular KPI.

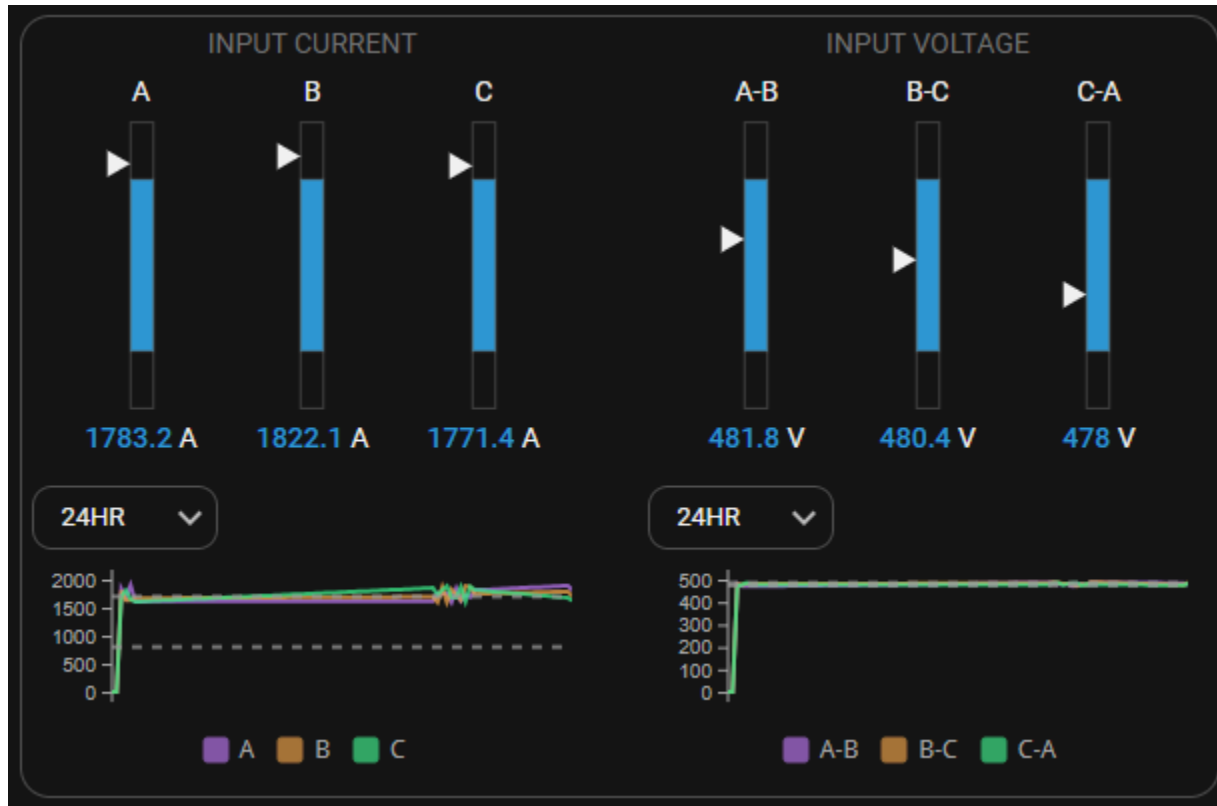
For each historical trend, we also display the minimum, maximum, and average values for the given time-period to provide further insight.

Parameters

Name	Description
tagPath	Path to the tag.
title	Title displayed at the top.

numberFormat	Format pattern of the value. The numberFormat expression function is used to format the value.
currentValueBasis	Flex-basis of the current value seen below the moving analog indicator. Default is auto. When set to auto, the width of the current value will change when the number of digits changes. If this behavior is not desirable, then this can be set to a numeric value.
rangesSelected	Settings for the time ranges selected. Possible values include 24HR, 7DAY, and 30DAY. • 0: Time range selected for the trend on top. Default is 24HR. • 1: Time range selected for the trend on the bottom. Default is 7DAY.
trend	Settings for the trends. • lineColor: Color of the trend line. • desired: Settings for the dashed desired range on the trends. ○ low: Low of the desired range. ○ high: High of the desired range.
movingAnalogIndicator	Moving Analog Indicator Parameters Note: the min-max range of the moving analog indicator comes from the engLow and engHigh of the tag.

Current & Voltage Widget



This **Current & Voltage Widget** displays current and voltage real-time values in moving analog indicators.

As long as the tags used with the widget are historized, trends will then be displayed showing 24 hour history to show at a glance what's been happening recently with the tags.

To use this widget for output current & voltages instead of input, you can simply alter the labels and point to different tags. Similarly, the labels at the top of each moving analog indicator can be edited to reflect any designated current or voltage.

This widget is also used in the UPS Popup that you can view by clicking the UPS Popup button below.

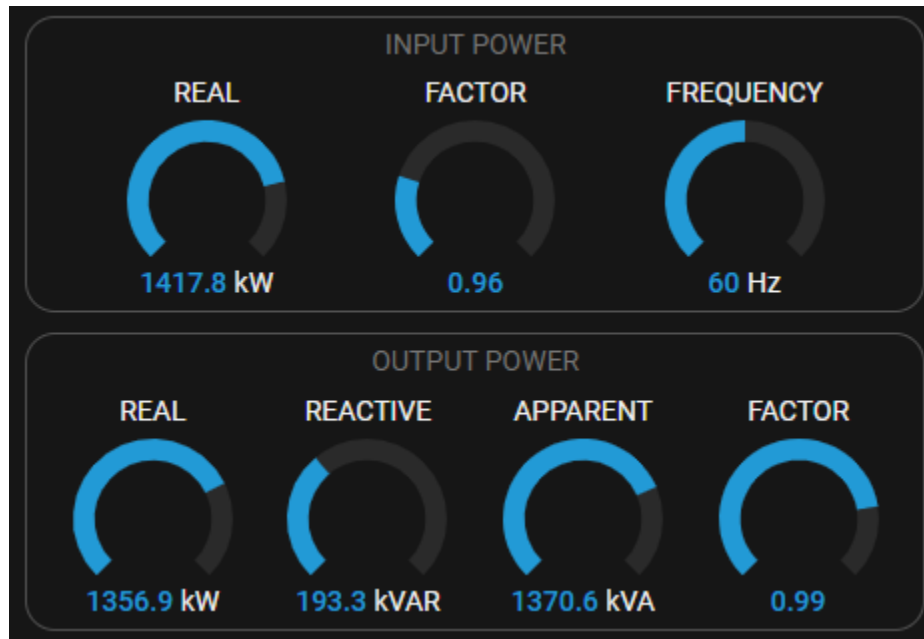
Parameters

Name	Description
tagPath	Path to the UDT or folder that contains the input power tags.

current	Settings for the current values. • title: Title displayed above the current moving analog indicators. • trend: Settings for the current trend. ○ numberFormat: Format pattern of the values in the trend. The numberFormat expression function is used to format the value. ○ rangeSelected: Time range selected in the trend. Possible values are 24HR, 7DAY, and 30DAY. Default is 24HR. ○ seriesName: Name of the series. ○ desired: Settings for the dashed desired range on the trends. ■ low: Low of the desired range. ■ high: High of the desired range. • values: Settings for the three current values. There are three objects titled: first (left value), second (middle value), and third (right value). Each of these objects contains: ○ pathToTag: Path to the tag that is located within the tag/folder at tagPath. ○ title: Title displayed above the moving analog indicator. ○ numberFormat: Format pattern of the value. The numberFormat expression function is used to format the value. ○ trend: Settings for the trend. ■ aggregate: Aggregation mode for the Tag History binding. ■ alias: Alias of the pen. ■ color: Color of the trend line. ○ maiBasis: Flex-basis of the moving analog indicator. ○ movingAnalogIndicator: Moving Analog Indicator Parameters Note: the min-max range of the moving analog indicator comes from the engLow and engHigh of the tag.
voltage	Settings for the voltage values. • title: Title displayed above the voltage moving analog indicators. • trend: Settings for the voltage trend. ○ numberFormat: Format pattern of the values in the trend. The

	numberFormat expression function is used to format the value. ○ rangeSelected: Time range selected in the trend. Possible values are 24HR, 7DAY, and 30DAY. Default is 24HR. ○ seriesName: Name of the series. ○ desired: Settings for the dashed desired range on the trends. ■ low: Low of the desired range. ■ high: High of the desired range. ● values: Settings for the three voltage values. There are three objects titled: first (left value), second (middle value), and third (right value). Each of these objects contains: ○ pathToTag: Path to the tag that is located within the tag/folder at tagPath. ○ title: Title displayed above the moving analog indicator. ○ numberFormat: Format pattern of the value. The numberFormat expression function is used to format the value. ○ trend: Settings for the trend. ■ aggregate: Aggregation mode for the Tag History binding. ■ alias: Alias of the pen. ■ color: Color of the trend line. ○ maiBasis: Flex-basis of the moving analog indicator. ○ movingAnalogIndicator: Moving Analog Indicator Parameters Note: the min-max range of the moving analog indicator comes from the engLow and engHigh of the tag.
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Input & Output Power Widgets



These **Input & Output Power Widgets** display power values in a set of simple gauges.

The **Input Power Widget** displays values for Real power, power Factor, and power Frequency.

The **Output Power Widget** displays values for Real power, Reactive power, Apparent power, and power Factor.

These widgets are also used in the UPS Popup that you can view by clicking the UPS Popup button below.

Parameters

Input Power

Name	Description
tagPath	Path to the UDT or folder that contains the input power tags.
title	Title displayed above the simple gauges.
simpleGauges	Settings for the three simple gauges. There are three objects titled: first (left value), second (middle value), and third (right value). Each of these objects

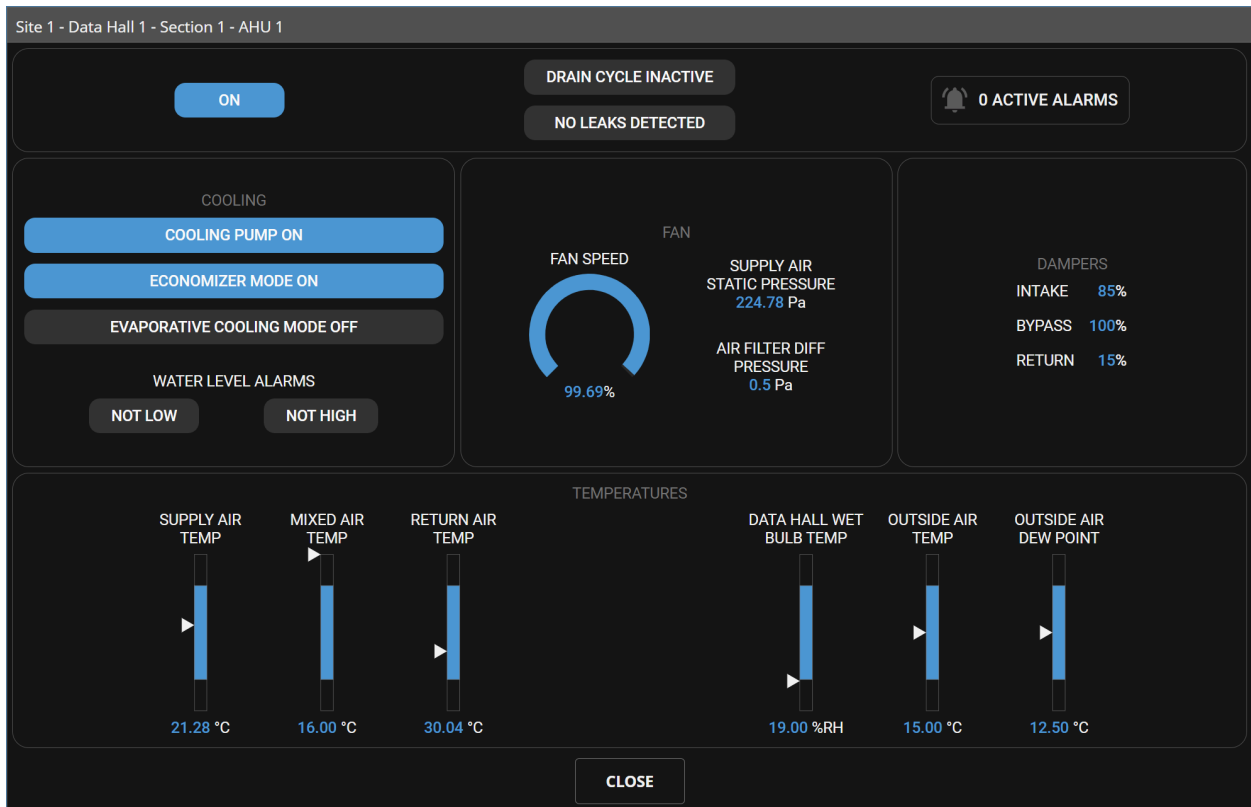
	contains: • pathToTag: Path to the tag that is located within the tag/folder at tagPath. • numberFormat: Format pattern of the value. The numberFormat expression function is used to format the value. • title: Title displayed above the simple gauge. • Simple Gauge Parameters Note: the min-max range of the simple gauge comes from the engLow and engHigh of the tag.
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Output Power

Name	Description
tagPath	Path to the UDT or folder that contains the output power tags.
title	Title displayed above the simple gauges.
simpleGauges	Settings for the four simple gauges. There are four objects titled: first (left value), second (middle-left value), third (middle-right value), and fourth (right value). Each of these objects contains: • pathToTag: Path to the tag that is located within the tag/folder at tagPath. • numberFormat: Format pattern of the value. The numberFormat expression function is used to format the value. • title: Title displayed above the simple gauge. • Simple Gauge Parameters Note: the min-max range of the simple gauge comes from the engLow and engHigh of the tag.

Equipment Popups / Faceplates

AHU



Click this button to view the **Air Handling Unit (AHU) Popup/Faceplate**. This popup displays general status for the unit, Cooling, Fan, and Damper KPIs, along with key temperature values.

While this Perspective view is being used as a popup here, it could also be used as an embedded view within another view as a dedicated, always-visible faceplate to display the AHU details without needing to open a popup.

Parameters

Name	Description
tagPath	Path to the UPS UDT/folder.

UPS



Click this button to view the **Uninterruptible Power Supply (UPS) Popup/Faceplate**. This popup shows general status for the unit, input & output values for power, current, and voltages, as well as battery status details.

While this Perspective view is being used as a popup here, it could also be used as an embedded view within another view as a dedicated, always-visible faceplate to display the UPS details without needing to open a popup.

Parameters

Name	Description
tagPath	Path to the AHU UDT/folder.

Dynamic

Site 1 - Data Hall 1 - Section 1 - UPS 1A			
☐ Active Time		Display Path	Priority Name
There are no active alarms			
Status	Battery	Output	
Active Alarm Count	0	System Output Apparent Power	1346.97 kVA
System Status String	Automatic	System Output Current A	1624.6 A
Is Inverter Active	True	System Output Current B	1644.8 A
Is Output Overloaded	False	System Output Current C	1591.1 A
System Status	1	System Output Current Mean	1620.2 A
Info		System Output Frequency	60 Hz
DataHall	1	System Output Power Factor	0.99
EquipmentId	1A	System Output Reactive Power	190.01 kVAR
EquipmentType	UPS	System Output Real Power	1333.5 kW
Section	1	System Output Voltage A-B	480 V
SerialNumber	SN123456	System Output Voltage B-C	480 V
Site	1	System Output Voltage C-A	480 V
Energy	Input	System Output Voltage Mean	480 V
System Input Cumulative			
Energy per Period	1200 GWh	System Input Current A	1790.5 A
System Output Cumulative		System Input Current B	1747.5 A
Energy per Period	997 GWh	System Input Current C	1866.4 A
		System Input Current Mean	1801.5 A
		System Input Frequency	60 Hz
		System Input Power Factor	0.97
		System Input Real Power	1455.68 kW
		System Input Voltage A-B	475.18 V
		System Input Voltage B-C	481.92 V
		System Input Voltage C-A	484.71 V
		System Input Voltage Mean	480.6 V
Close			

Click this button to view the **Dynamic Popup**. With this popup, you set a base tag folder path, which will then browse the tags within that folder, and display all of the sub tags, showing the tag name, real-time value, and engineering units.

Note: This popup is optimized for on the order of 50 tags. If you would like to display more than this number, simply make a copy of this popup, resize it, and point the tag path at this new larger popup.

For this example, the tag path is set to a single instance of a UPS UDT.

Parameters

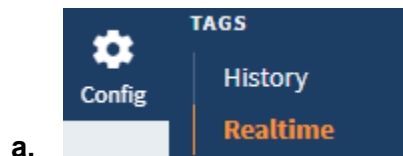
Name	Description
tagPath	Path to the UDT.
mobile	If true, the layout of the information is optimized for mobile.

Tags and UDTs

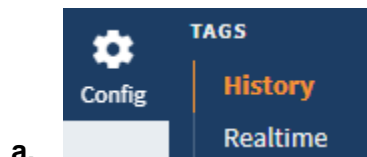
Tags can vary greatly depending on equipment type and manufacturer. Included tags are a template that point to a Programmable Device Simulator. Users might need to modify these to suit their needs.

Setup

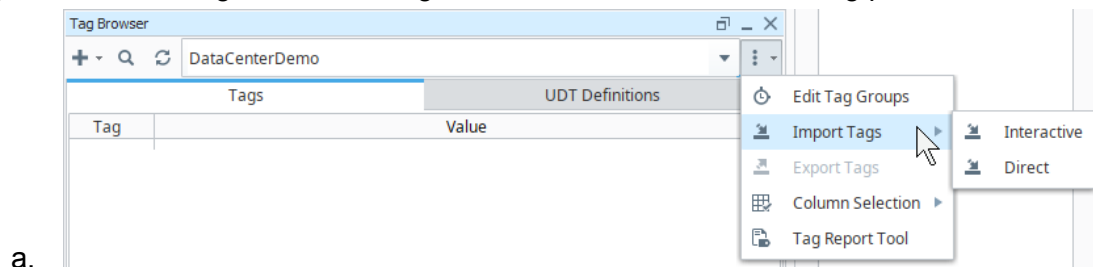
1. In the Gateway config Tags > Realtime page create a realtime tag provider named **DataCenterDemo**



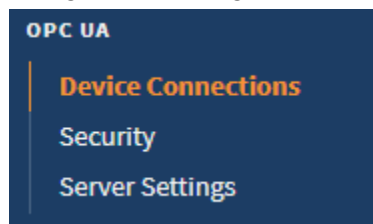
2. In the Gateway config Tags > Historical page create a historical tag provider named **DataCenter**



3. Import UDTs and tags into the Designer, under **DataCenterDemo** tag provider



4. (Optional) In the Gateway config OPC UA > Device Connections page, Create a new Programmable Device Simulator named **DataCenterDemoOpc**. If mapping to a different device this step may be skipped.
 - a. Navigate to Config > OPC UA > Device Connections



- b. Click Create new Device, select Programmable Device Simulator, name it DataCenterDemoOpc and Create New Device

Name	Type
DataCenterDemoOpc	Programmable Device Simulator

→ Create new Device...

- c. After creation click More and Edit Program

Name	Type	Description	Enabled	Status	
DataCenterDemoOpc	Programmable Device Simulator		true	Running	More edit

→ Create new Device...

edit program

delete

- d. In the Load Program choose Load from CSV, Choose File and browse to the included csv file, then click Load Simulator Program and Save Program. The csv file is located in the Tags folder.

DataCenterDemoOpc

Load Program

1

Load from CSV

2

Choose File

No file chosen

3

Load Simulator Program

Export Instruction File

Clear All

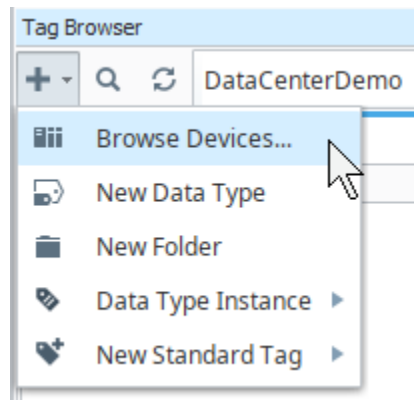
Time Interval	Browse Path	Value Source	Data Type
Add Instruction			

4

Save Program

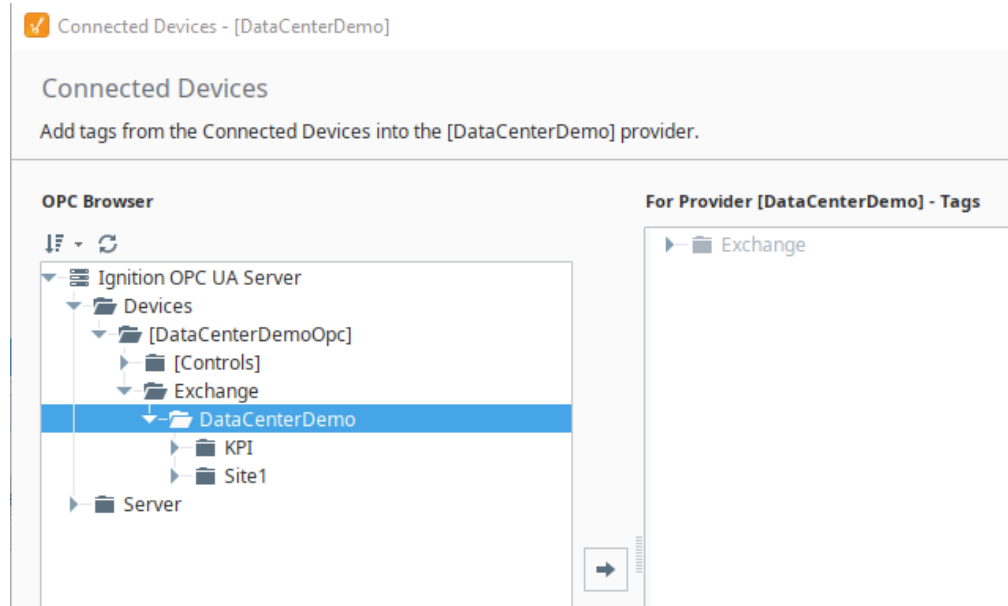
<< >> >>>

- e. To validate the setup browse devices in the Designer



i.

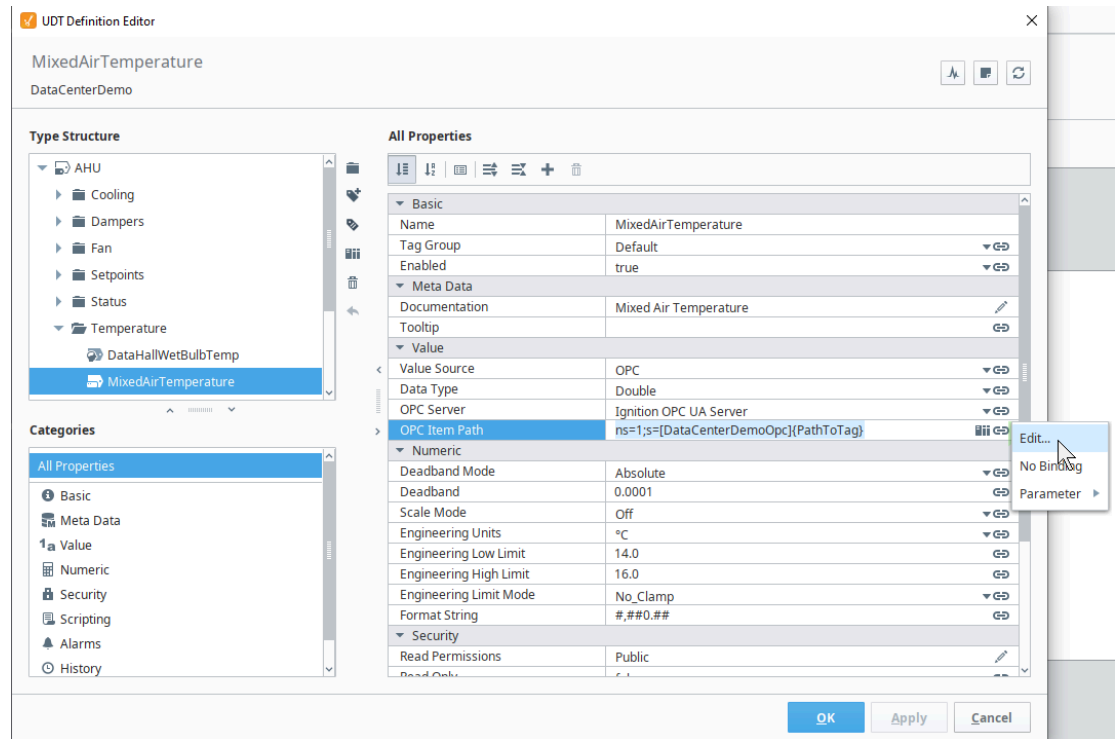
- ii. You should be able to see DataCenterDemo folder



iii.

- iv. It might take up to a minute for tags to start updating

5. (Optional) Connecting to different equipment. This step can be skipped if the tags are not being connected to actual equipment and are used for demonstration purposes only.
 - a. Change OPC Item Path for each tag within the UDT to the desired OPC path by browsing or edit the binding



b.

AHU

The Air Handling Unit is used for temperature control.

UDT Path:

[DataCenterDemo]_types_/Exchange/DataCenterDemo/Equipment/AHU

Sample Tag Path:

[DataCenterDemo]Exchange/DataCenterDemo/Site1/DataHall1/Section1/AHU/AHU1

AHU1		
Parameters		
Cooling		
IsCoolingPumpOn		☒
IsCoolingWaterLevelHigh		☐
IsCoolingWaterLevelLow		☐
IsEvaporativeCoolingActive		☐
IsUnitInEconomizerMode		☒
Dampers		
BypassDamperFeedback	100	
BypassDamperPosition	100	
OutsideAirDamperFeedback	85	
OutsideAirDamperPosition	85	
ReturnAirDamperFeedback	15	
ReturnAirDamperPosition	15	
Fan		
FanSpeedCommand	70	
FanSpeedFeedback	74.57	
IsFanAlarmActive		☐
SupplyAirStaticPressure	213.27	
Setpoints		
SupplyAirTemperature	20	
Status		
ActiveAlarmCount	0	
AirFilterDifferentialPressure	0.5	
IsAhuRunning		☒
IsDrainCycleActive		☐
IsLeakDetected		☐
StartCommand		☒
Temperature		
DataHallWetBulbTemp	19	
MixedAirTemperature	14.76	
OutsideAirDewPoint	12.5	
OutsideAirTemperature	15	
ReturnAirTemperature	31.09	
SupplyAirTemperature	21.27	
Info		

UPS

Uninterruptible Power Supply is used as a backup power supply.

UDT Path:

[DataCenterDemo]_types_/Exchange/DataCenterDemo/Equipment/UPS

Sample Tag Path:

[DataCenterDemo]Exchange/DataCenterDemo/Site1/DataHall1/Section1/UPS/UPS1A

UPS1A Parameters Battery BatteryCapacityRemaining BatteryDischargeCycles ElapsedTimeDischargingBattery IsBatteryCharging IsBatteryDischarging IsBatteryLow IsBatteryTempHigh IsBatteryTestFailed IsBatteryTestInProgress Energy SystemInputRealEnergyMonthly SystemOutputRealEnergyMonthly Input SystemInputCurrentA SystemInputCurrentB SystemInputCurrentC SystemInputCurrentMean SystemInputFrequency SystemInputPowerFactor SystemInputRealPower SystemInputVoltageAB SystemInputVoltageBC SystemInputVoltageCA SystemInputVoltageMean Output Status Info		98.72 100 0 ☒ ☐ ☐ ☐ ☐ ☐ 1,200 997 1,733.68 1,845.54 1,686.16 1,755.13 60 0.97 1,407.09 472.65 481.79 480.51 478.32
UPS1A Parameters Battery Energy Input Output SystemOutputApparentPower SystemOutputCurrentA SystemOutputCurrentB SystemOutputCurrentC SystemOutputCurrentMean SystemOutputFrequency SystemOutputPowerFactor SystemOutputReactivePower SystemOutputRealPower SystemOutputVoltageAB SystemOutputVoltageBC SystemOutputVoltageCA SystemOutputVoltageMean Status ActiveAlarmCount ControlSwitchStatusString IsInverterActive IsOutputOverloaded SystemStatus Info		1,349.69 1,676.82 1,624.17 1,569.28 1,623.43 60 0.99 190.4 1,336.2 480 480 480 480 0 Automatic ☒ ☐ 1

KPI

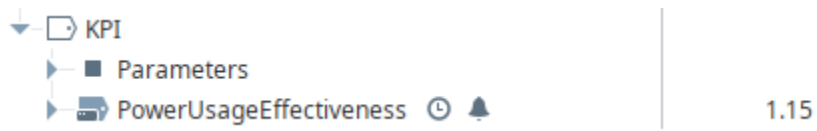
Key Performance Indicators UDT contains a placeholder PUE tag. More tags can be added here per user's specifications.

UDT Path:

[DataCenterDemo]_types_/Exchange/DataCenterDemo/KPI

Sample Tag Path:

[DataCenterDemo]Exchange/DataCenterDemo/KPI



Info

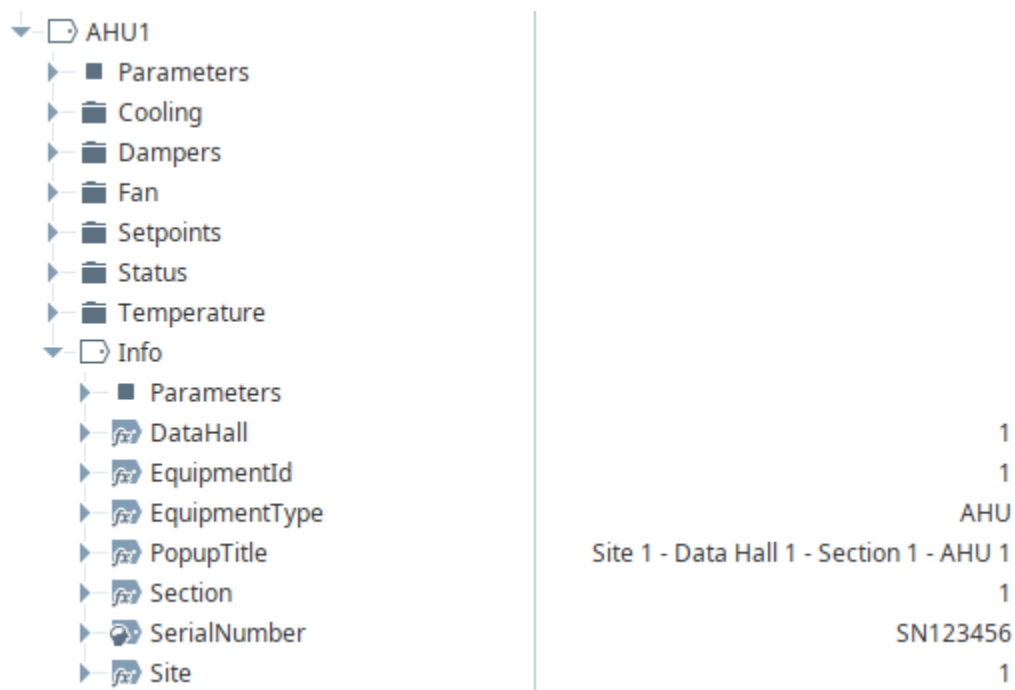
This UDT is embedded in other equipment UDTs for a standardized manner of displaying location data. It parses the tag path to derive site, data hall, and section of the parent UDT. Then creates a PopupTitle tag with a user friendly text.

UDT Path:

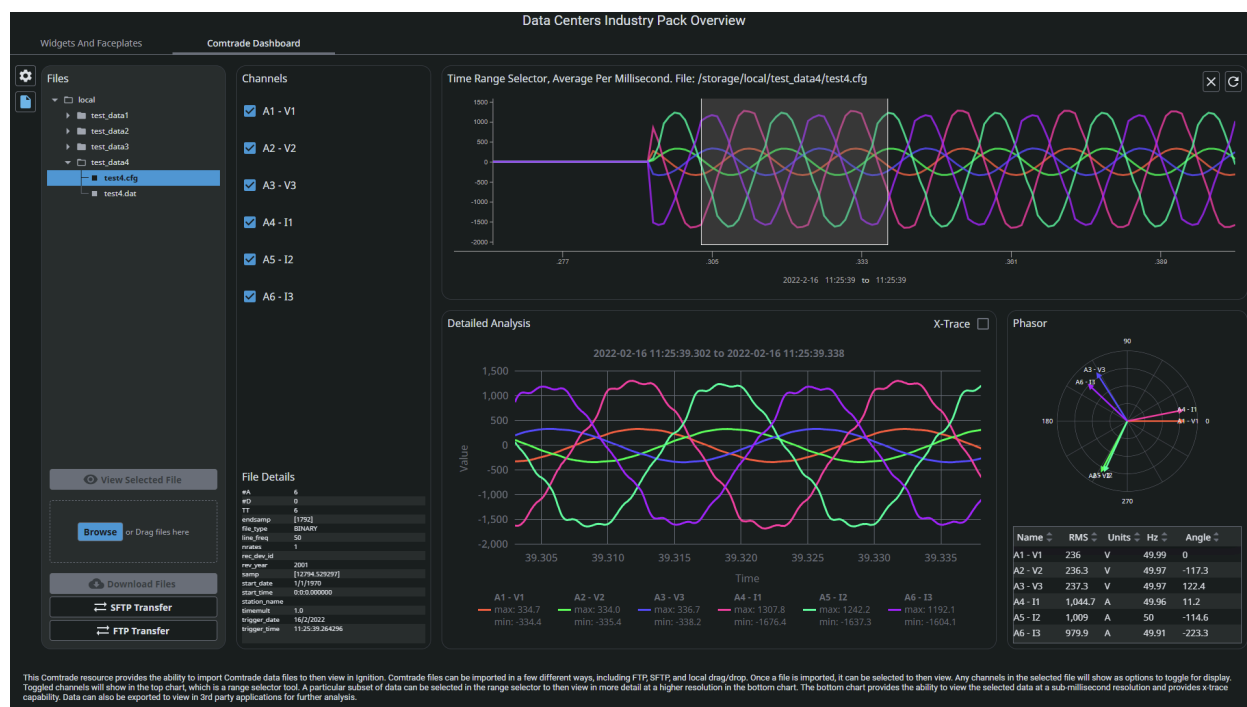
[DataCenterDemo]_types_/Exchange/DataCenterDemo/Info

Sample Tag Path:

[DataCenterDemo]Exchange/DataCenterDemo/Site1/DataHall1/Section1/AHU/AHU1/Info



Comtrade



This Comtrade resource provides the ability to import Comtrade data files to then view in Ignition. Comtrade files can be imported in a few different ways, including FTP, SFTP, and local drag/drop.

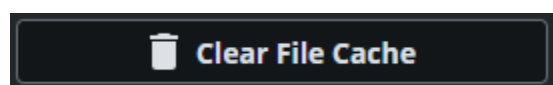
Once a file is imported, it can be selected to then view. Any channels in the selected file will show as options to toggle for display. Toggled channels will show in the top chart, which is a range selector tool.

A particular subset of data can be selected in the range selector to then view in more detail at a higher resolution in the bottom chart. The bottom chart provides the ability to view the selected data at a sub-millisecond resolution and provides x-trace capability.

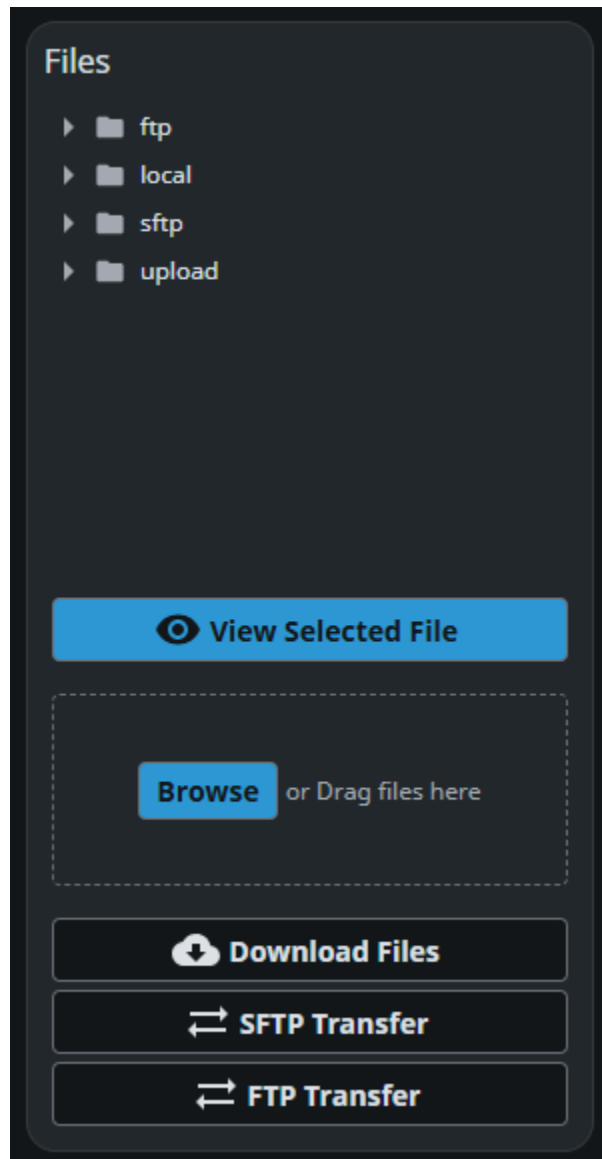
Data can also be exported to view in 3rd party applications for further analysis.

Cache

For performance considerations this resource utilizes global variables to store parsed Comtrade data for each session and page. Click **Clear File Cache** when data is no longer needed. Please note this will remove all cached Comtrade data from the Gateway for all sessions. If a more robust and/or automatic cache management system is needed it should be implemented by the end user.



File management



Tree - Displays files already present on the Gateway. Select a file from the tree and click **View Selected File** to parse and show the data.

View Selected File - Visualize the selected file.

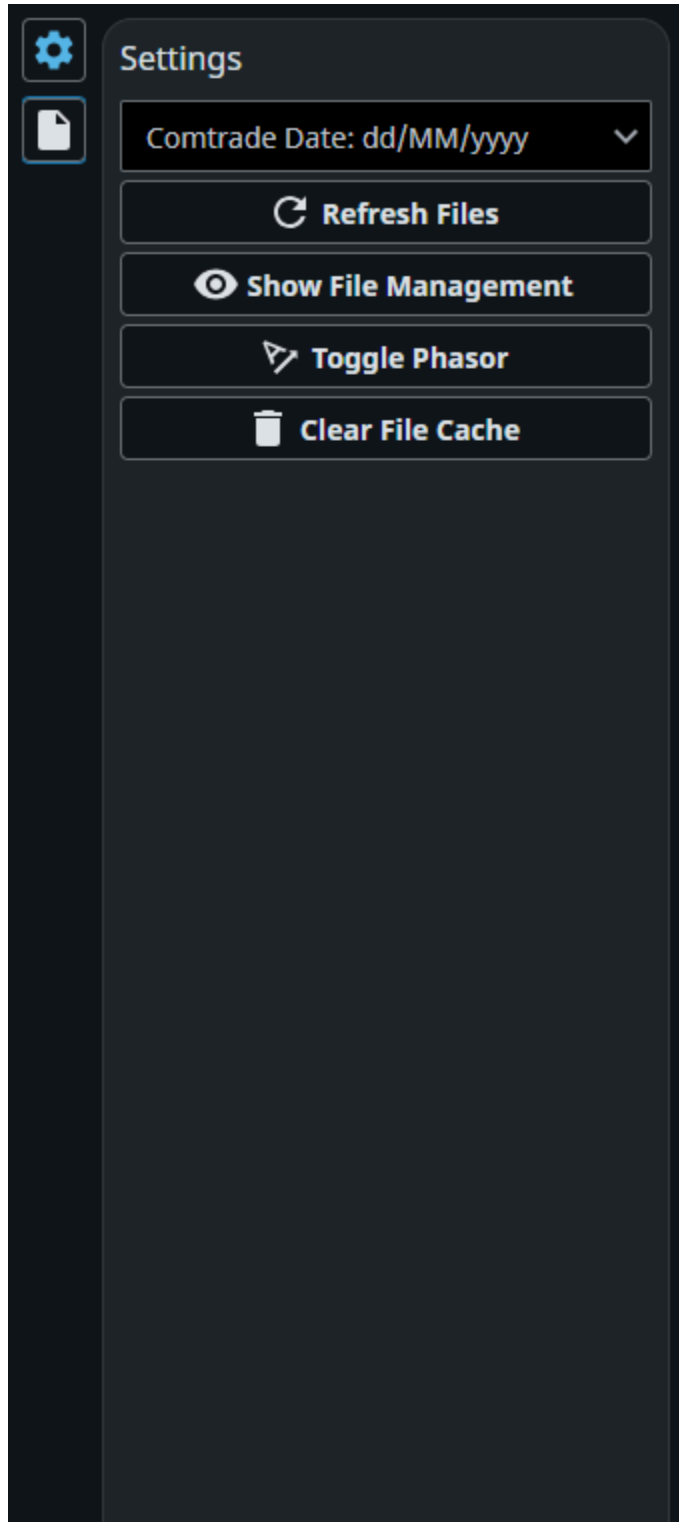
Upload - Drag in Comtrade files, or browse to upload files to the Gateway. Files will be placed in the upload folder. Uploading at least **.cfg** and **.dat** files is required to parse the data.

Download Files - Gateway will zip **.cfg .dat .inf** files and start download.

SFTP Transfer - This will transfer files over SFTP, placing them into the **sftp** folder. (Requires additional setup)

FTP Transfer - This will transfer files over FTP, placing them into **ftp** folder. (Requires additional setup)

Settings



Comtrade Date - Date format used inside of the Comtrade file. Once changed click **View Selected File** to apply format change.

Refresh Files - Refreshes file list, this action happens automatically when loading the view.

Show File Management - Shows or hides file management buttons.

Toggle Phasor - Shows or hides phasor component.

Clear File Cache - Clears cached Comtrade data from the Gateway. Note this removes the Comtrade data from all sessions, see [Cache](#) for more info.

File Details

Channels

☒ A1 - V1

☒ A2 - V2

☒ A3 - V3

☐ A4 - I1

☐ A5 - I2

☐ A6 - I3

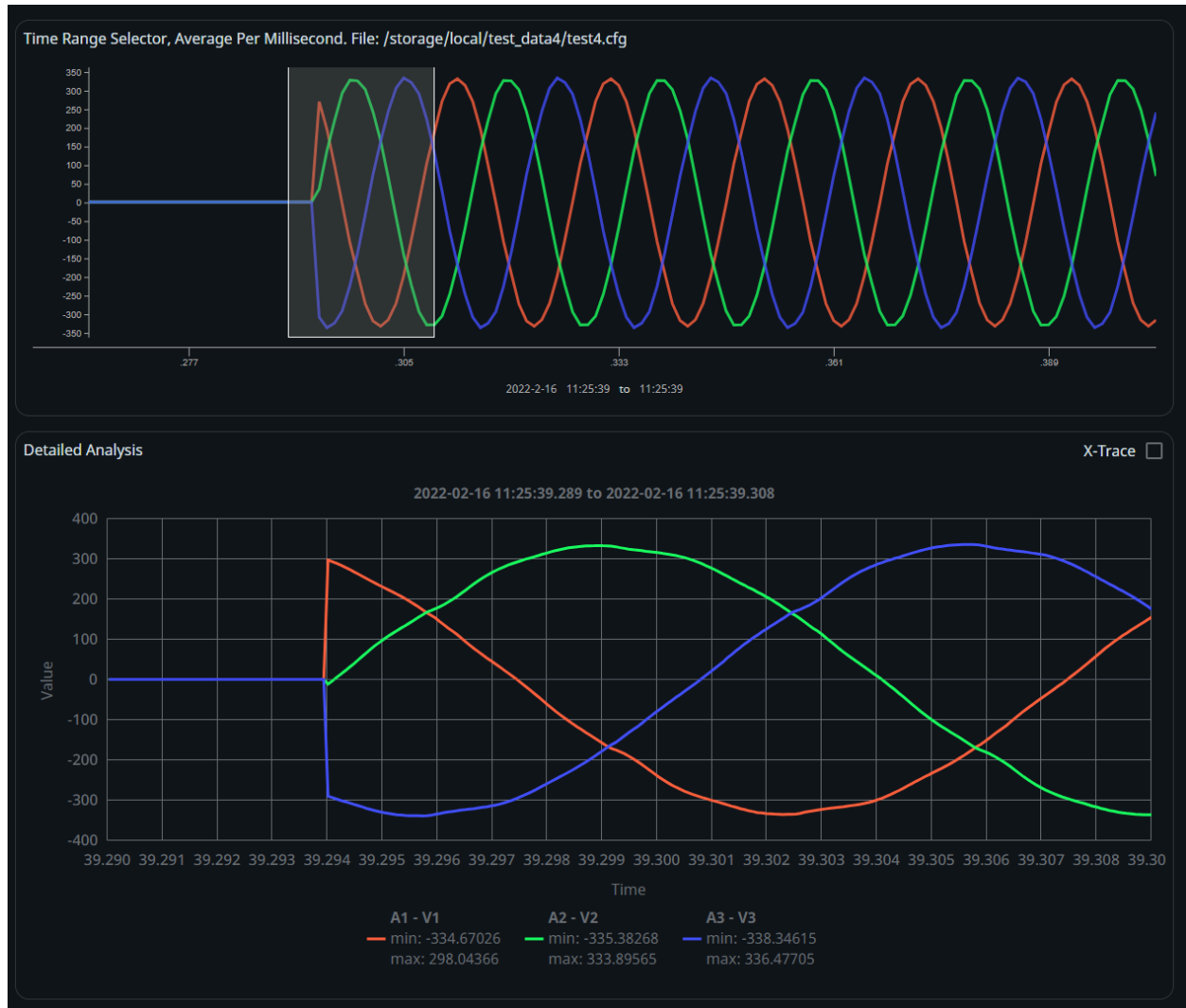
File Details

#A	6
#D	0
TT	6
endsamp	[1792]
file_type	BINARY
line_freq	50
nrates	1
rec_dev_id	
rev_year	2001
samp	[12794.529297]
start_date	1/1/1970
start_time	0:0:0.000000
station_name	
timemult	1.0
trigger_date	16/2/2022
trigger_time	11:25:39.264296

Channels - This shows a list of available channels in the Comtrade file. Checkbox toggles visibility of the channel in the charts.

File Details - Shows meta data from the configuration file, including timestamps and number of channel types.

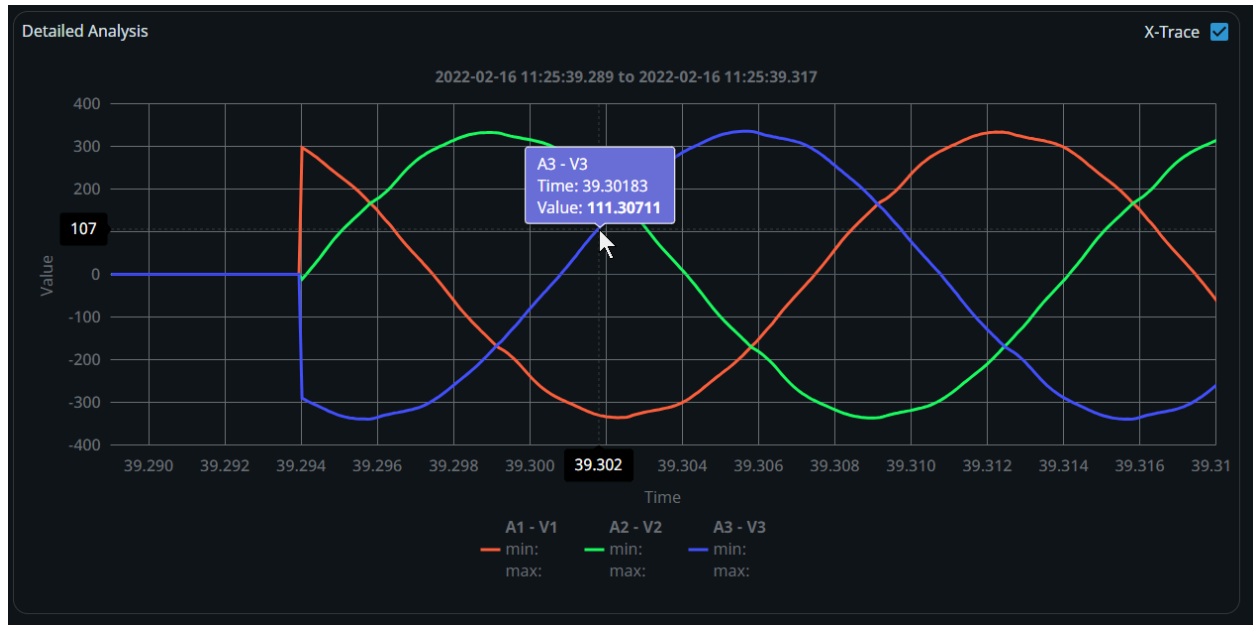
Time Range Selector



The top graph shows aggregated data for selected channels. Values are averaged by millisecond, selecting a time range in this component will show raw data in the bottom chart.

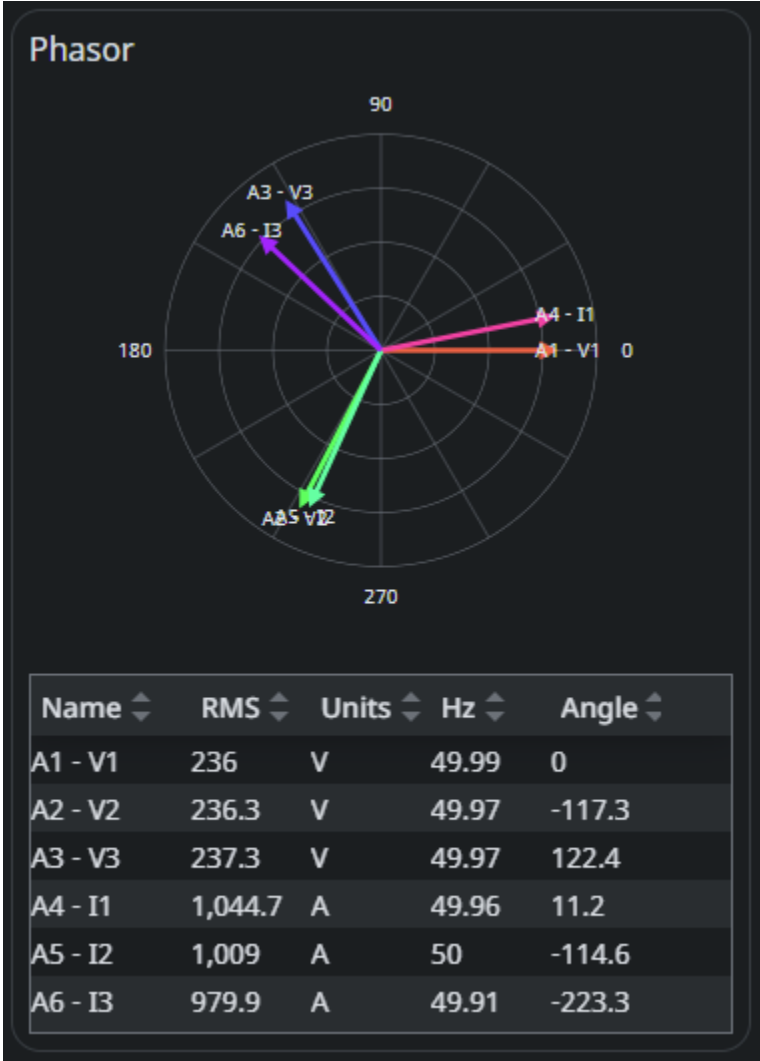
Detailed Analysis

X-Trace



Clicking the **X-Trace checkbox** in the top right of the component will load additional data for every point along the chart.

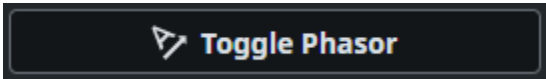
Phasor



The phasor is calculated for the selected time range. Selecting a specific row in a table will rotate the graph to align it to 0 degrees.

The phasor is calculated by Fast Fourier Transform that is then fed into a Sine fitting function.

Visibility of this component is toggled by the **Toggle Phasor button** in **settings (gear icon)**.

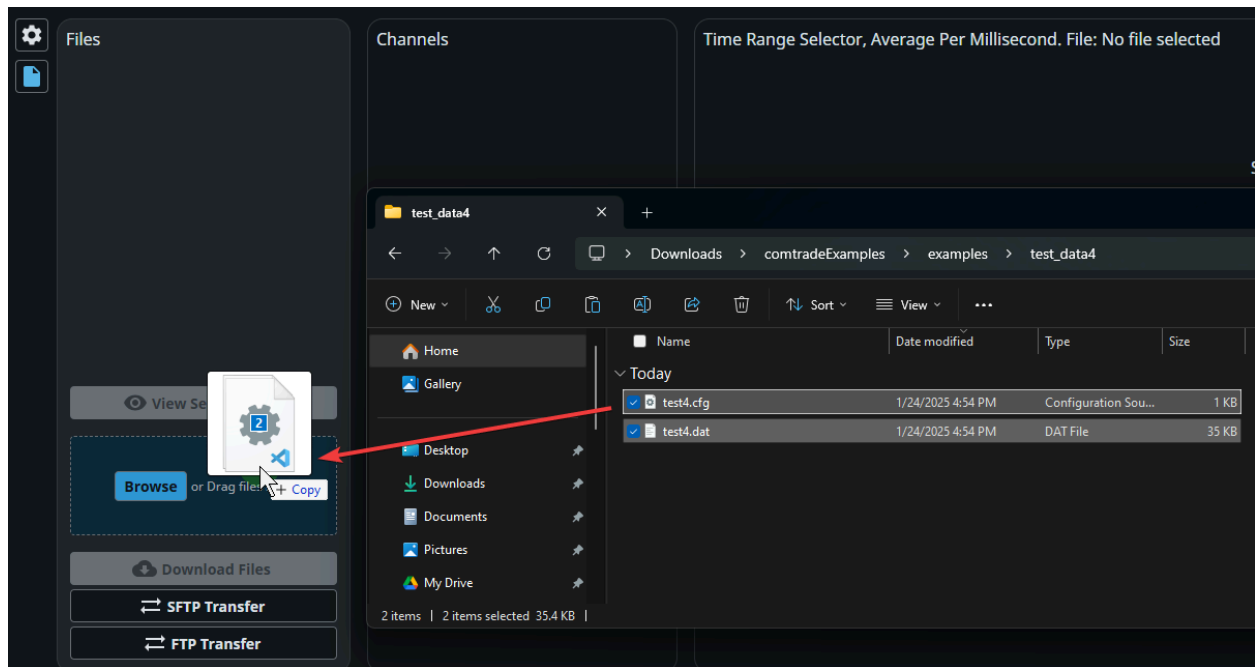


Comtrade Setup

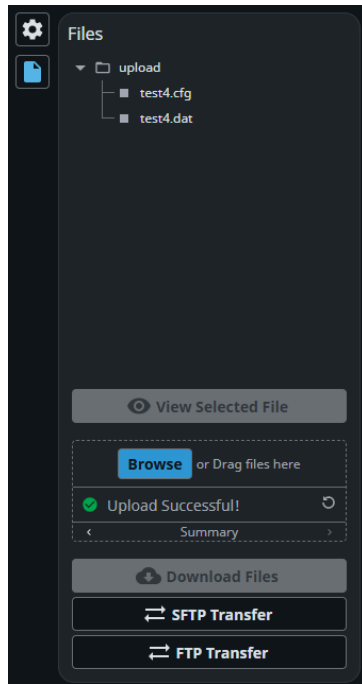
Upload

Perform this test first, it will check basic functionality of Comtrade parsing. If successful you can move onto FTP and/or SFTP setup if desired.

1. Locate and extract contents of **ComtradeExamples.zip** file.
2. Browse into one of the examples and drag them into upload section of the resource
 - a. Using **test_data4** in this example.



3. Once the files are uploaded you should see the **upload** folder.



4. Refresh files if you do not see the **upload** folder. Then select the file, click **View Selected File**, and select a time range in the **Range Selector**.



5. If you see data plotted in the charts, then the upload test was successful.

FTP

Note the following FTP steps can be ignored if FTP is not required to import your Comtrade data.

An FTP server or device is required for this step.

Ignition contains ftplib which allows connecting to FTP servers directly from scripting.

```
Python
import ftplib
```

Example implementation of downloading files over FTP:

```
Python
downloadFtpFiles(ftpHost='myFTP', ftpPort=21, ftpUser='ftp', ftpPass='ftp',
remoteDir='/Comtrade/', localDir=exchange.comtrade.utils.folderFtp())
```

The following information must be configured per device by the end user:

ftpHost='myFTP' (hostname or IP address of FTP server)

ftpPort=21 (port that is used for transfer, default is 21)

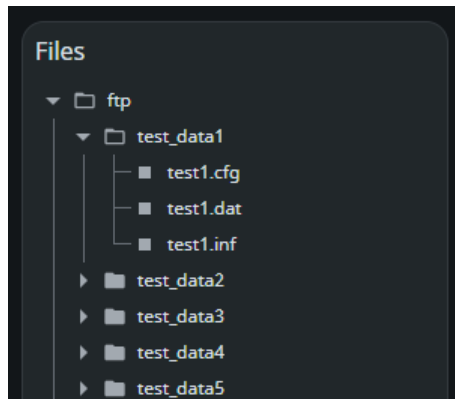
ftpUser='ftp' (username)

ftpPass='ftp' (password)

remoteDir='/Comtrade/' (remote directory to copy from FTP server, make sure this folder exists and there are Comtrade files inside, **ComtradeExamples** files are available as part of this resource)

localDir=exchange.comtrade.utils.folderFtp() (directory where files will be copied to)

Once FTP Transfer is clicked an **ftp folder** with appropriate files should appear.



SFTP

Note the following SFTP steps can be ignored if SFTP is not required to import your Comtrade data.

An SFTP server or device is required for this step.

This resource was developed on a Linux Gateway, other operating systems are not currently supported but the same techniques can be used to develop similar scripts for any operating system.

To accomplish this transfer Ignition triggers the **sftpcopy.sh** shell script, which transfers files from a server onto the Gateway. Follow these steps to setup:

1. Install following Linux packages on the Gateway

None

```
apt-get install -y openssh-client sshpass
```

2. Update **SFTP server details** with credentials of SFTP server in **sftpcopy.sh**
 - a. This file is included inside of **scripts.zip**

```
1  #!/bin/bash
2
3  # SFTP server details
4  SFTP_USER="sftp"
5  SFTP_PASS="sftp"
6  SFTP_HOST="mySFTP"
7  REMOTE_DIR="/comtrade"
8  LOCAL_DIR="/usr/local/bin/ignition/webserver/webapps/comtrade/storage/sftp"
9
10 # Create the local directory if it does not exist
11 mkdir -p $LOCAL_DIR
12
13 # Create a temporary file to store SFTP commands
14 SFTP_COMMANDS=$(mktemp)
15
16 # Write SFTP commands to the temporary file
17 echo "lcd $LOCAL_DIR" > $SFTP_COMMANDS
18 echo "cd $REMOTE_DIR" >> $SFTP_COMMANDS
19 echo "mget -r *" >> $SFTP_COMMANDS
20
21 # Run SFTP with the commands from the temporary file using sshpass
22 sshpass -p $SFTP_PASS sftp -oBatchMode=no -oStrictHostKeyChecking=no -b $SFTP_COMMANDS $SFTP_USER@$SFTP_HOST
23
24 # Clean up the temporary file
25 rm $SFTP_COMMANDS
```

3. Move modified **sftpcopy.sh** script to the following path on the Gateway:
/usr/local/bin/ignition/webserver/webapps/comtrade/scripts/sftpcopy.sh
4. Allow execution of the script by running the following command in the folder of the Gateway where you just placed the script.

None

```
sudo chmod +x sftpcopy.sh
```

5. Test if the script is working by triggering it manually.

None

```
./sftpcopy.sh
```

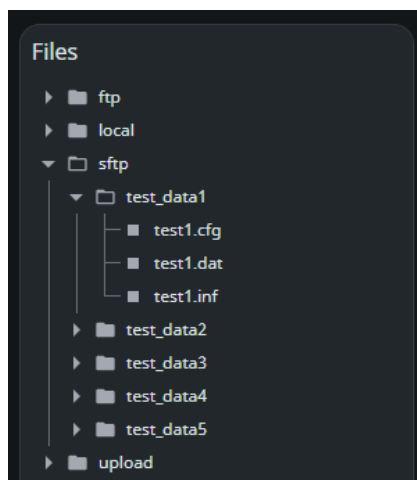
6. If the script finished without errors press **Refresh Files**, you should now see the **sftp folder** with files transferred from the SFTP server. Note: this requires files on the SFTP server. If there aren't any files, you can use the included **ComtradeExamples** files for testing.

 **Refresh Files**

7. To verify that triggering from Ignition functions as expected either delete files from **/usr/local/bin/ignition/webserver/webapps/comtrade/storage/sftp** or add new files to the SFTP server and click the **SFTP Transfer** button. Double check that the files appear in the tree component. Clicking the **Refresh Files** button might be required if transfer took more than half a second.

 **SFTP Transfer**

 **Refresh Files**



Additional Resources

Simple Gauge



This resource provides a way to show a real-time value in a range as it changes. It has just one axis, is easy to configure and is customizable in its appearance.

It is used in the **Input Power** and **Output Power Widgets**, where the process value is retrieved from the tag at the tagPath parameter, and the min-max range comes from the tag's engLow and engHigh properties.

If it is desirable to specify the process value and min-max range manually rather than retrieving these values from the tag via bindings, then the view with the following viewPath can be used instead: *Exchange/SimpleGauge/SimpleGauge*.

Name	Description						
startAngle	Radial position for the start of the gauge's arc.						
endAngle	Radial position for the end of the gauge's arc.						
arc	Radial band that displays the gauge's value. The arc properties: <table><tr><th>Name</th><th>Description</th></tr><tr><td>width</td><td>Width of the arc in the SVG coordinate system. Must be greater than 0.</td></tr><tr><td>color</td><td>Color of the arc line (as hexadecimal) showing the gauge's value.</td></tr></table>	Name	Description	width	Width of the arc in the SVG coordinate system. Must be greater than 0.	color	Color of the arc line (as hexadecimal) showing the gauge's value.
Name	Description						
width	Width of the arc in the SVG coordinate system. Must be greater than 0.						
color	Color of the arc line (as hexadecimal) showing the gauge's value.						
arcBackground	Background or 'track' for the gauge arc. The arc background properties: <table><tr><th>Name</th><th>Description</th></tr><tr><td>color</td><td>Color of the arc background (as hexadecimal).</td></tr></table>	Name	Description	color	Color of the arc background (as hexadecimal).		
Name	Description						
color	Color of the arc background (as hexadecimal).						

	<table> <tr> <td>opacity</td><td>Opacity of the arc. 0 is fully transparent, 1 is fully opaque.</td></tr> </table>	opacity	Opacity of the arc. 0 is fully transparent, 1 is fully opaque.														
opacity	Opacity of the arc. 0 is fully transparent, 1 is fully opaque.																
label	Label that displays the value of the gauge. The label properties: <table> <tr> <th>Name</th><th>Description</th></tr> <tr> <td>visible</td><td>Indicates whether or not the label is visible.</td></tr> <tr> <td>size</td><td>Font size to display label text.</td></tr> <tr> <td>offsetX</td><td>Offset position on the x axis, relative to the middle of the gauge.</td></tr> <tr> <td>offsetY</td><td>Offset position on the y axis, relative to the middle of the gauge.</td></tr> <tr> <td>color</td><td>Color of the label.</td></tr> <tr> <td>units</td><td>Any unit information to append to the value.</td></tr> <tr> <td>maxDecimal</td><td>Maximum number of digits after decimal.</td></tr> </table>	Name	Description	visible	Indicates whether or not the label is visible.	size	Font size to display label text.	offsetX	Offset position on the x axis, relative to the middle of the gauge.	offsetY	Offset position on the y axis, relative to the middle of the gauge.	color	Color of the label.	units	Any unit information to append to the value.	maxDecimal	Maximum number of digits after decimal.
Name	Description																
visible	Indicates whether or not the label is visible.																
size	Font size to display label text.																
offsetX	Offset position on the x axis, relative to the middle of the gauge.																
offsetY	Offset position on the y axis, relative to the middle of the gauge.																
color	Color of the label.																
units	Any unit information to append to the value.																
maxDecimal	Maximum number of digits after decimal.																

Moving Analog Indicator



This resource displays an analog value in context with other information about that value so that you can quickly see if the value is in the normal range or not.

It is used in the **KPI Details** and **Current & Voltage Widgets**, where the process value is retrieved from the tag at the tagPath parameter, and the min-max range comes from the tag's engLow and engHigh properties.

If it is desirable to specify the process value and min-max range manually rather than retrieving these values from the tag via bindings, then the view with the following viewPath can be used instead: *Exchange/SVGMovingAnalogIndicator/SVGMovingAnalogIndicator*.

Name	Description
defaultRangeColor	Color of the default range, which spans from minValue to maxValue.
desired	Settings for the desired range. • low: Lower limit of the desired range. • high: Upper limit of the desired range. • rangeColor: Color of the desired range.
alarms	Settings for the low, high, low-low, and high-high alarm ranges. • level1: Settings for the level 1 alarms (low-low or high-high). ○ lowLow: Value below which is a low-low alarm. ○ highHigh: Value above which is a high-high alarm. ○ rangeColor: Color of the active level 1 alarm. • level2: Settings for the level 2 alarms (low or high).

	○ low: Value below which is a low alarm. ○ high: Value above which is a high alarm. ○ rangeColor: Color of the active level 2 alarm. ● inactiveRangeColor: Color of the inactive color range.
interlocks	Settings for the interlock ranges. ● low: Value below which an interlock will be activated. ● high: Value above which an interlock will be activated. ● rangeColor: Color of the interlock ranges.
setpoint	Settings for the setpoint value marker. ● value: Current value of the setpoint. ● color: Color of the setpoint value marker. ● stroke: Settings for the stroke of the setpoint value marker. ○ color: Color of the stroke. ○ width: Width of the stroke.
label	Settings for the process value label. ● visible: Whether to display the label. ● format: Format pattern of the value. The numberFormat expression function is used to format the value. ● position: Settings for the label position. ○ horizontalOffset: Offset label position horizontally. ○ verticalOffset: Offset label position vertically. ○ applyOffsetsOnlyWhenLabelVisible: If true, the offsets will be applied only if the label is visible. If false, then the offsets will always be applied regardless of whether or not the label is visible.

	Note about offsets: If <code>label.visible=True</code> or <code>label.applyOffsetsOnlyWhenLabelVisible=False</code>, then a nonzero horizontal offset when <code>verticalDisplay=True</code> or a nonzero vertical offset when <code>verticalDisplay=False</code> influences the position of the indicator and the ranges bar. If it is desirable to be able to toggle the label visibility without affecting the position of the indicator and the ranges bar, then <code>applyOffsetsOnlyWhenLabelVisible</code> should be set to <code>False</code>. • <code>style</code>: Style of the label. Properties shared between CSS and SVG can be used. ○ <code>font-size</code>: Size of the label. ○ <code>fill</code>: Color of the label.
<code>rangesBar</code>	Settings for the ranges bar. • <code>size</code>: Size of the ranges bar. • <code>verticalDisplay</code>: Display the ranges bar vertically. • <code>stroke</code>: Settings for the stroke of the ranges bar. ○ <code>color</code>: Color of the stroke. ○ <code>width</code>: Width of the stroke.
<code>indicator</code>	Settings for the process value indicator. • <code>size</code>: Size of the indicator. • <code>color</code>: Color of the indicator. • <code>reverse</code>: Displays the indicator on the opposite side of the ranges bar. • <code>stroke</code>: Settings for the stroke of the indicator. ○ <code>color</code>: Color of the stroke. ○ <code>width</code>: Width of the stroke.